

Rwanda Electricity Access Trends

2009 – 2024 · Data Analytics Project
Personal Data Analytics Project · April 2026

75%	84.6%	+380%	100%
Household access (2024 — World Bank)	Total connectivity (Jul 2025 — REG)	Relative growth (2010–2020)	Health centers electrified

1. Introduction

Rwanda has achieved one of the fastest electricity access expansions in Africa, transforming its electrification rate from just 6% of households in 2009 to 75% by March 2024 — a 15-year journey that has been recognised globally. This report examines the data behind that progress, analyses the urban–rural access gap, quantifies year-on-year growth rates, and identifies the policy milestones that drove acceleration. The analysis is directly relevant to the renewable energy sector: as of July 2025, 25% of Rwanda's 84.6% household connectivity comes from off-grid solar systems — representing a significant and rapidly growing market for off-grid energy businesses across Rwanda.

2. National Access Trend (2009–2023)

The World Bank SDG 7.1.1 dataset shows Rwanda's national access rate rising from 6.0% (2009) to 63.9% (2023), a gain of 57.9 percentage points over 14 years. The table below summarises the data, including year-on-year changes:

Year	Access (%)	YoY Change (pp)	Note
2009	6.0	—	Baseline — Vision 2020 launched
2010	9.8	+3.8	EDPRS I in progress
2011	10.0	+0.2	—
2012	10.8	+0.8	—
2013	11.5	+0.7	—
2014	19.4	+7.9	EDPRS II begins; REG established
2015	26.0	+6.6	—
2016	34.0	+8.0	Grid expansion acceleration
2017	39.2	+5.2	—
2018	43.0	+3.8	Off-grid SHS mandate
2019	46.6	+3.6	—
2020	45.2	–1.4	COVID-19 — only decline recorded
2021	48.7	+3.5	Post-pandemic recovery

Year	Access (%)	YoY Change (pp)	Note
2022	50.6	+1.9	—
2023	63.9	+13.3	Largest single-year jump on record

Key finding: The largest single-year gain (+13.3 pp) was recorded in 2023, suggesting accelerating momentum. The only negative year was 2020 (–1.4 pp), reflecting disruptions from the COVID-19 pandemic.

3. Urban–Rural Access Gap

Rwanda's 86% rural population creates a structural challenge: urban areas benefited from grid proximity first, creating a persistent access gap. However, the data shows this gap has begun to narrow significantly, driven by off-grid solar programmes targeting remote households:

Year	Urban Access (%)	Rural Access (%)	Gap (pp)	Trend
2010	58.0	5.0	53	Gap widening initially
2014	68.0	12.0	56	EDPRS II rural push begins
2018	75.0	17.0	58	Peak gap — off-grid mandate introduced
2022	87.0	40.0	47	Gap narrows — SHS programmes scaling

The rural access rate grew 8× (from 5% to 40%) between 2010 and 2022, compared to 1.5× urban growth. This acceleration reflects the government's dual-track strategy: grid extension for denser areas and subsidised Solar Home Systems for remote households.

4. Grid vs Off-Grid Breakdown (2025)

As of July 2025, Rwanda Energy Group (REG) reports a total household connectivity rate of 84.6%, comprising two distinct channels:

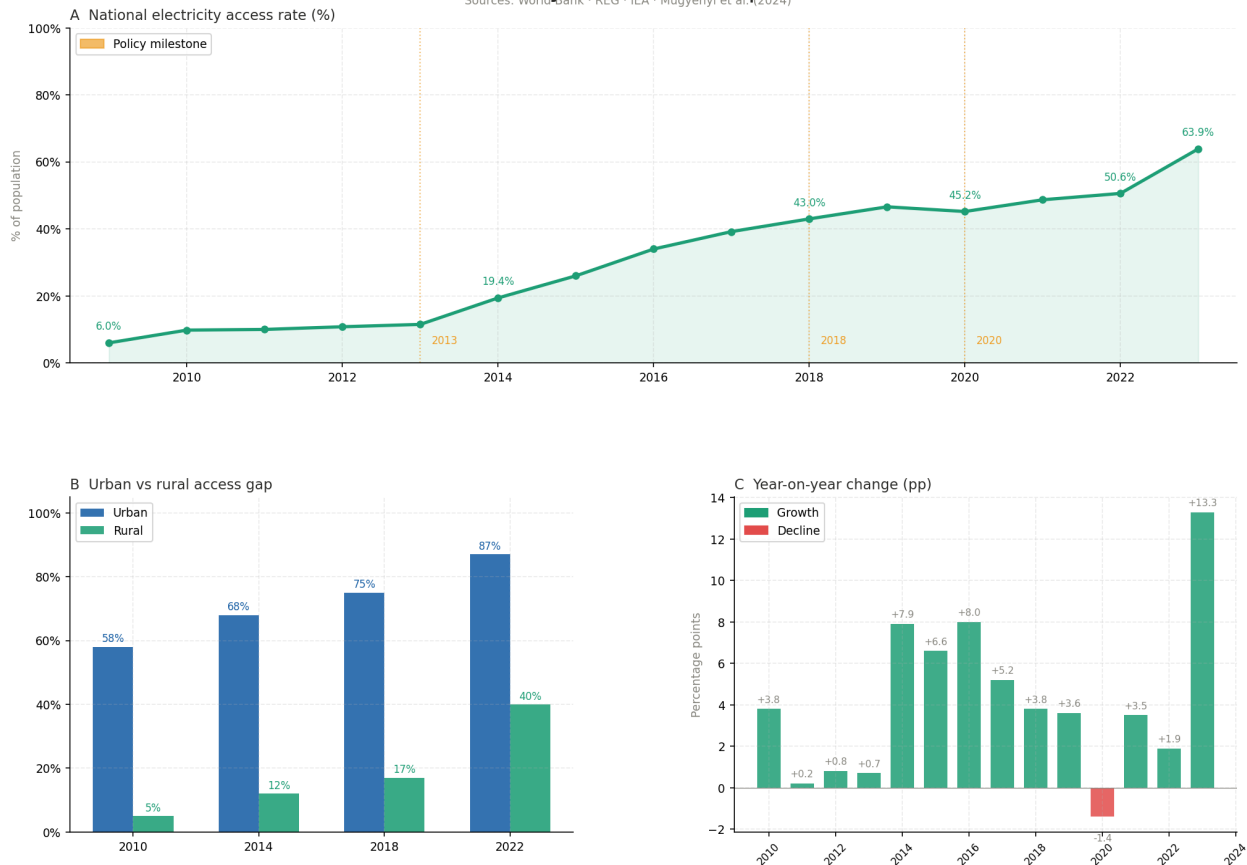
Connection Type	Share (%)	Households	Sector relevance
On-grid (national grid)	59.6%	~2.0M	Grid expansion opportunities
Off-grid solar (SHS/mini-grid)	25.0%	~850K	Core solar product market
No access (remaining gap)	15.4%	~520K	Target for 2030 universal goal

5. Visualisations

The chart below (generated from Python using pandas and matplotlib) summarises the three main dimensions of this analysis: national trend, urban–rural gap, and year-on-year growth rates.

Rwanda Electricity Access Trends | 2009 - 2024

Sources: World Bank · REG · IEA · Mugenyi et al. (2024)



6. Key Findings and Implications

Rwanda's electrification was the 3rd fastest in Africa (2010–2020)

ranking 11th globally with a 380% relative increase. This positions Rwanda as a model for renewable energy deployment in Sub-Saharan Africa.

The off-grid solar sector is maturing rapidly

25% of all connected households use Solar Home Systems — representing approximately 850,000 households and a significant commercial opportunity for renewable energy companies.

A 15.4% gap remains toward the 2030 universal access target

~520,000 households still lack electricity, concentrated in remote rural areas where off-grid solar is more cost-effective than grid extension.

COVID-19 had a measurable but temporary impact

The only negative year in the dataset was 2020 (–1.4 pp). Recovery was swift, and 2023 recorded the largest ever annual gain (+13.3 pp).

Data Sources

- World Bank SDG 7.1.1 Electrification Dataset (ESMAP / trackingsdg7.esmap.org)
- Rwanda Energy Group (REG) — Access Dashboard, July 2025 (reg.rw/what-we-do/access)
- International Energy Agency (IEA) — Rwanda Country Profile (iea.org/countries/rwanda)
- Mugenyi J. et al. (2024). 'Rwanda's Path to Universal Electricity Access: Consumption Trends, Tariff Impact, and Challenges Ahead.' SSRN. Columbia University QSEL.

- World Bank Feature Report (April 2024). 'Ingredients for Accelerating Universal Electricity Access: Lessons from Rwanda's Inspirational Approach.'

Analysis conducted in Python (pandas, matplotlib, numpy). Data accurate as of April 2026. All figures from official sources.